Muhammad Abdullah

syedabdullahjami@gmail.com ❖ (+92) 3159209292 ❖ Islamabad, Pakistan

LinkedIn: Muhammad-Abdullah GitHub: Muhammad-Abdullah-Jami

SKILLS

ML/DL Frameworks:	PyTorch, TensorFlow, Keras, Scikit-Learn, LangChain, LangGraph, AI Agents
Core AI Concepts:	Deep Learning, Machine Learning, Computer Vision, Natural Language Processing , Data Science, Digital Image Processing
Languages:	Python, C, C++, JavaScript, SQL
Libraries & Tools:	NumPy, Pandas, Matplotlib, OpenCV, NLTK, Selenium, BeautifulSoup, Asyncio,Tqdm, Pydub
Databases:	MySQL, MongoDB, PostgreSQL, Pinecone, PGVector
Model Deployment:	Docker, Jenkins (CI/CD), Ollama (Local LLM Framework), n8n Automation

Work Experience

AI Developer | *Zigron* | DentaSmart **Jan 2025 – Present**

- **Architected Multi-Agent Conversational Systems:**
  Engineered an advanced AI Chat and Voice ecosystem using LangGraph, FastAPI, and MySQL/PostgreSQL. Implemented sophisticated state management and recursive flow control to handle complex, multi-turn clinical dialogues.

- **Developed High-Performance RAG Pipeline:**
  Integrated pgvector for high-dimensional semantic search, optimizing document retrieval speed and accuracy. Engineered a custom context-aware summarization pipeline to maintain long-term memory across extended user sessions.

- **Engineered Real-Time Voice Infrastructure:**
  Designed and deployed a low-latency AI voice assistant utilizing WebSockets and Google Gemini. Achieved seamless, sub-second response times by optimizing the full duplex communication bridge between the client and server.

- **Developed Modular Speech-to-Speech Pipelines:**
  Built a specialized voice bot using the OpenAI Platform, architecting a custom STT → LLM → TTS pipeline.
  - **Speech-to-Text:** Optimized Whisper with domain-specific prompting and Client-side VAD to eliminate hallucinations.
  - **Intelligence:** Leveraged GPT-4o for high-context clinical reasoning.
  - **Text-to-Speech:** Integrated high-fidelity synthesis for natural-sounding responses, ensuring a seamless end-to-end user experience.

- **Implemented Enterprise-Grade Security:**
  Fortified the AI API infrastructure by implementing JWT (JSON Web Token) authentication, ensuring secure, stateless session management and protected access to sensitive data across MySQL and PostgreSQL databases.

- **Advanced Workflow Automation:**
  Built complex, data-driven AI workflows using n8n, Webhooks, and custom SQL integrations. Developed a recursive summarization pipeline that enables long-context memory retention for extended user sessions.

- **Directed AI DevOps:**
  Spearheaded the containerization and deployment of the entire AI stack using Docker and Jenkins (CI/CD), maintaining a robust pipeline for rapid iteration and 99.9% system availability.

Python Intern | Stocks Telegraph July 2023 – Aug 2023

- Managed Stocks data using the Django framework.

EDUCATION

National University of Computer and Emerging Sciences - FAST Aug 2019 – Dec 2024

BS, Artificial Intelligence Islamabad, Pakistan

PROJECTS

Automated Textile Defects Detection System using Computer Vision (Final Year Project)

- Developed a Computer Vision system using YOLOv8 and the TILDA dataset for defect classification.
- Collaborated with industry to integrate a custom dataset and deployed the solution as a web application.

Kage no Koe (Voice of Shadow) Local Mind

- Research-focused project exploring the capabilities of local, open-source Large Language Models (LLMs) i.e. DeepSeek, Qwen, Llama, the project leverages the Ollama framework with React based UI to replicate paid subscription like GPT, Claude into free of cost open-source local running chatbot.

Comparative Analysis for EV Price Prediction

- Developed a system comparing Linear Regression, Random Forest, XGBoost, and SVM.
- XGBoost achieved an R2 score of 0.90.

Stock Price Forecasting using Machine Learning

- Data Built a forecasting system comparing LSTM, Random Forest, XGBoost, and SVR
- Random Forest achieved R2 scores over 0.99.

Time Series Modeling and Forecasting

- Compared statistical (ARIMA) and deep learning (RNN, GRU, LSTM) models for stock price forecasting
- Found ARIMA provided precise predictions based on trend, while LSTM captured generalized trends.

Transfer Learning for Image Classification

- Compared VGG16 and ResNet on a food dataset
- Demonstrated that ResNet with unfrozen layers provided superior performance

Data Mining and Classification

- Applied Decision Tree, Random Forest, and custom ANN to the Breast Cancer dataset.
- Determined Random Forest was the most effective classifier.

Certifications

- *Python CLI Chat Application using LangChain* — Built AI-powered CLI chat apps with LangChain, LLMs, memory integration, and agent-based workflows (Udemy, Jan 2026)
- *Practical Prompt Engineering for Work (Tool-Agnostic*) — Applied structured prompt design techniques to improve LLM outputs across tools and real-world use cases (Udemy, Dec 2025)